A convergence theorem in profinite equivariant stable homotopy theory

with a quick peek at Chromatic Homotopy Theory and tt-geometry

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Two convergence results

Theorem (Hopkins–Ravenel, Chromatic Convergence, Ravenel 1992, 7.5.7)

For a finite spectrum X, its p-localisation $X_{(p)}$ can be computed as the limit of its $E(n)$-localisations:

$$X_{(p)} \simeq \lim_n L_n X.$$

Proposition ("Geometric Convergence" in Sp_G, H.)

Let G be a top. fin. gen. pro-p-group. For a uniformly bounded below genuine G-spectrum X, its "geometric localisation" $L^{\mathrm{geo}} X$ can be computed as a certain limit of cofibers of assembly maps:

$$L^{\mathrm{geo}} X \simeq \lim_{\substack{N \trianglelefteq G \\ \text{open normal}}} \widetilde{\mathbf{E}\mathcal{P}_N} \otimes X.$$

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Lethal amount of hand-waving, homotopy theory and algebra.



From abelian groups to spectra

- (Connective) **spectra** to spaces are like abelian groups to sets.
- "non-$\mathbb{Z}$-linear" linear algebra, generalised homology theories..
- X pointed space: $\pi_1(X) = \pi_0(\Omega X)$ group, $\pi_2(X) = \pi_0(\Omega^2 X)$ abelian
- "Loop groups" $\Omega^n X$ become **more and more** commutative as n grows.

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From abelian groups to spectra

Definition (Spectra := infinite loop space)

A spectrum X is a sequence $X_n : \mathrm{An}_*$, $n \in \mathbb{Z}_{\geq 0}$ of pointed anima spaces, together with equivalences $X_n \simeq \Omega X_{n+1}$. They constitute a stable ∞-category Sp.

- $\pi_n(X) \in \mathrm{Ab}$ for $n \in \mathbb{Z}$ as $\pi_{n+q}(X_q)$ for $q \gg 0$
- sphere spectrum $\mathbb{S}$, $\pi_\bullet(\mathbb{S})$ stable homotopy groups of spheres
- $\mathbb{S}$ the initial commutative ring spectrum, spectrum = "derived $\mathbb{S}$-module".

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"tt" stands for tensor triangular, triangular ≈ stable.

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monoidal product	$\otimes_{\mathbb{Z}}^{(\mathbb{L})}$ tensor product	$\otimes_{\mathbb{S}}$ smash product
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geometric object	Zariski spectrum	(Balmer) tt-spectrum

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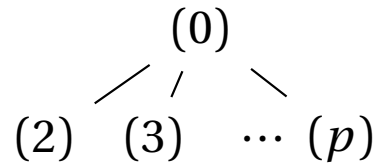
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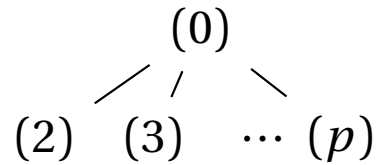
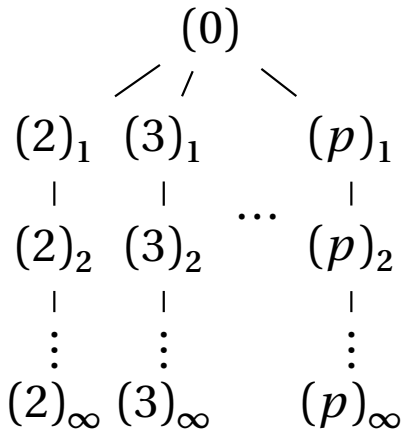
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What are $(p)_h$ and why Ravenel called them "chromatic"?

- Primes of $\mathbb{S} \leftrightarrow$ points of a certain stack $\mathcal{M}_{\mathrm{fg}}$ of (graded) formal groups.

$$(p)_h \leftrightarrow \text{the formal group of height } h \text{ at prime } p$$

(nilpotence and periodicity theorem (Devinatz)–Hopkins–Smith))

prime point:	(0)	$(p)_\infty$	$(p)_h$ for $0 < h < \infty$
"residue field":	$\mathbb{Q}$	$\mathbb{F}_p$	Morava K-theory $K(h)$
	discrete		$K(h)_\bullet \simeq \mathbb{F}_p[v_h^{\pm}], \lvert v_h \rvert = 2p^h - 2$

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- $(p)_h$ related to periodicity phenomena in Sp..

.. (Ravenel:) of different wave length degree $= 2p^h - 2$

Open subsets of $\mathrm{Spec}(\mathbb{S}_{(p)})$

$$\underbrace{(0) - (p)_1 - (p)_2 - \cdots - (p)_n}_{\text{open subset } U_{\leq n}} - \cdots (p)_\infty$$

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- Restriction to $U_{\leq n}$ a "smashing localisation" $L_n^{\mathrm{f}}(-) \simeq L_n^{\mathrm{f}}\mathbb{S} \otimes (-)$

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- For $X \in \mathrm{Sp}$: $L_0^{\mathrm{f}} X = L_0 X = X \otimes \mathbb{Q}$
  $L_1^{\mathrm{f}} = L_1 =$ Bousfield localisation at $\mathrm{KU}_{\hat{p}}$
  $L_n^{\mathrm{f}} \neq L_n$ for $n \geq 2$ (Burklund–Hahn–Levy–Schlank)

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- $\exists$ chromatic tower $X \to \ldots \to L_n X \to L_{n-1} X \to \ldots \to L_0 X$

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Theorem (Hopkins–Ravenel, Chromatic Convergence, Ravenel 1992, 7.5.7)

For a finite spectrum X, its chromatic tower converges to $X_{(p)} \simeq \lim_n L_n X$.

"We can recover X from the 'harmonic part' $X|_{U_{<\infty}}$."
(Convergence of the tower consisting of $L_n^{\mathrm{f}} X$ is still open.)

Spectra of genuine G-spectra (Balchin, Barnes and Barthel 2024)

- $\mathrm{Spec}(\mathrm{Sp}_G) \simeq \coprod_{\substack{H \leq G \\ \text{closed}}} \mathrm{Spec}(\mathbb{S})$ as a set, topology partially understood
- open subset U_N on $H \supseteq N$ ($N \trianglelefteq G$ open) a smashing localisation Φ^N
- $U_{\mathrm{fin}} := \bigcup_N U_N$: "geometric localisation & tower" $L^{\mathrm{geo}} \Rightarrow \ldots \Rightarrow \Phi^N$,

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Theorem ("Geometric convergence", H.)

Let G be a profinite group w/ finiteness condition (e.g. top. fin. gen. pro-p-group). For a uniformly bounded below G-spectrum X, its geometric tower converges to its geometric localisation:

$$L^{\mathrm{geo}} X \simeq \lim_{\substack{N \trianglelefteq G \\ \text{open normal}}} \Phi^N X.$$

"We can recover X from the 'finite index part' $X|_{U_{\mathrm{fin}}}$."

References I


Scott Balchin, David Barnes and Tobias Barthel (2024). Profinite equivariant spectra and their tensor-triangular geometry.


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